

TECHNICAL REPORT: Data Architecture & Statistical Methodology

1. Data Engineering & Pipeline Architecture

The primary hurdle in this analysis was resolving mismatched Primary Keys between the CRM (customer_churn.csv) and POS (sales_data.csv) databases.

- **Transformation:** We utilized Pandas string manipulation (.str.replace) to strip alphabetical prefixes ('C00001' and 'CUST001') and cast the remaining strings as integers.
- **Merging:** A LEFT JOIN was executed on the CRM dataset. This preserved the required 500-row baseline while appending transactional data.
- **Imputation:** Customers without hardware purchases returned NaN values post-join. These were mathematically imputed using fillna(0) to allow for accurate continuous variable testing.

2. Exploratory Data Analysis (EDA)

EDA was conducted using Seaborn to visualize multidimensional relationships:

- **Categorical Risk:** Countplots revealed distinct churn variations based on the type of product purchased.
- **Financial Density:** Scatterplots sizing transactions by Quantity visualized the clustering of high-revenue customers, highlighting that high Monthly Charges act as a ceiling for retention.
- **Multicollinearity:** A Pearson Correlation Heatmap quantified the financial relationships between Tenure, Monthly Charges, and Total Sales.

3. Inferential Statistical Validation

To prevent business strategies from relying on observational charts alone, strict parametric testing was deployed using scipy.stats. An alpha threshold of $\alpha = 0.05$ was established.

- **Hypothesis 1 (Hardware vs. Subs-Only):** An Independent T-Test was run comparing the churn means of hardware buyers versus non-buyers. The result ($p < 0.05$) successfully rejected the Null Hypothesis, proving hardware ecosystem integration significantly reduces churn.
- **Hypothesis 2 (Monthly Charges vs. Churn):** A second T-Test compared the mean monthly charges of churned vs. retained users. The result ($p < 0.05$) rejected the Null Hypothesis, validating "bill shock" as a primary attrition driver.

4. Conclusion

The pipeline successfully transitions raw, siloed data into actionable intelligence. The statistical significance of the findings guarantees that the proposed 3-phase implementation plan (Hardware subsidization and bill-shock prevention) is mathematically sound and primed for high ROI.